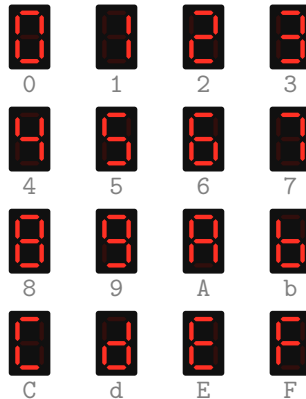
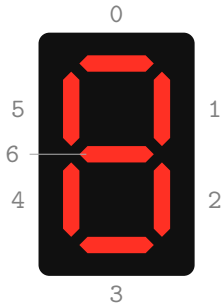


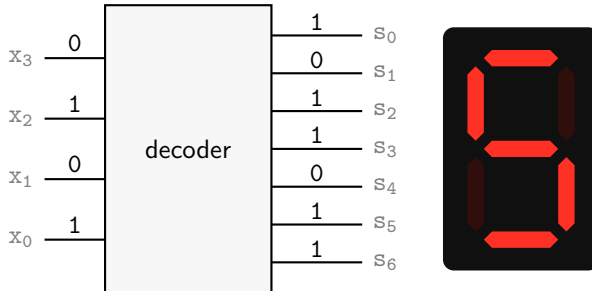
seven-segment decoders

A **seven-segment display** draws a digit by lighting some of seven LEDs, numbered 0 to 6. Different patterns give all sixteen hex digits.



b and d are lower case, so they aren't mistaken for 8 and 0

A **seven-segment decoder** reads a 4-bit number and outputs the seven segment signals, s_0 to s_6 , that draw it.



input 0101 lights segments 0, 2, 3, 5 and 6, drawing a 5

Each segment is its own function of the four inputs. Segment 0 is lit for every digit except 1, 4, b and d.

segment 0

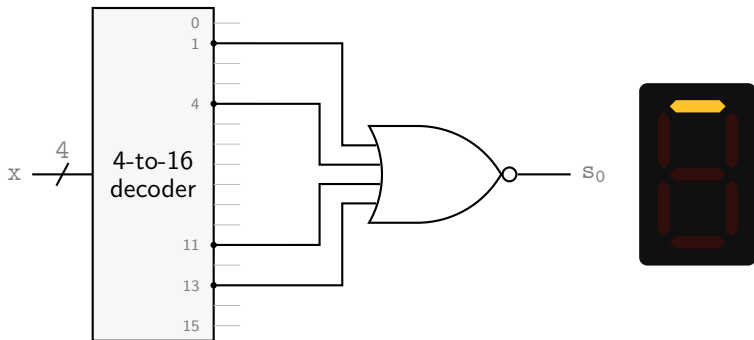


X		S ₀
0000	0	1
0001	1	0
0010	2	1
0011	3	1
0100	4	0
0101	5	1
0110	6	1
0111	7	1

X		S ₀
1000	8	1
1001	9	1
1010	A	1
1011	b	0
1100	c	1
1101	d	0
1110	e	1
1111	f	1

the other six segments each get a table like this one

Segment 0 is off in only four rows, so a **decoder** builds it directly: we NOR the outputs for 1, 4, 11 and 13.



a digit display for 0–9 only would make rows 1010–1111
don't-cares